

Oscillator strengths and lifetimes

In Chapter 3 we considered briefly the photoexcitation of Rydberg atoms, paying particular attention to the continuity of cross sections at the ionization limit. In this chapter we consider optical excitation in more detail. While the general behavior is similar in H and the alkali atoms, there are striking differences in the optical absorption cross sections and in the radiative decay rates. These differences can be traced to the variation in the radial matrix elements produced by nonzero quantum defects. The radiative properties of H are well known, and the radiative properties of alkali atoms can be calculated using quantum defect theory.

Oscillator strengths

A convenient way of expressing the strengths of transitions is to use the oscillator strength. The oscillator strength $f_{n'\ell'm',n\ell m}$ from level $n\ell m$ to level $n'\ell'm'$ is defined by¹

$$f_{n'\ell'm',n\ell m} = 2 \frac{m}{\hbar} \omega_{n'\ell',n\ell} |\langle n'\ell'm' | x | n\ell m \rangle|^2, \quad (4.1)$$

where $\omega_{n'\ell',n\ell} = (W_{n'\ell'} - W_{n\ell})/\hbar$. We could equally well have used the matrix elements of y or z in Eq. (4.1), but in any case, $f_{n'\ell'm',n\ell m}$ is m dependent. Since it is evident that the radiative decay of an atom in free space cannot depend upon m , it is useful to introduce the average oscillator strength $\bar{f}_{n'\ell',n\ell}$ which does not depend upon m . It may be written as¹

$$\bar{f}_{n'\ell',n\ell} = \frac{2}{3} \omega_{n'\ell',n\ell} \frac{\ell_{\max}}{2\ell+1} |\langle n'\ell' | r | n\ell \rangle|^2, \quad (4.2)$$

where $\ell_{\max}$ is the larger of ℓ and ℓ' . If we reverse the roles of ℓ and ℓ' , it is straightforward to show that

$$\bar{f}_{n'\ell',n\ell} = -\frac{2\ell'+1}{2\ell+1} \bar{f}_{n\ell,n'\ell'}. \quad (4.3)$$

The usefulness of the oscillator strengths stems in part from the fact that they satisfy several sum rules. The Thomas–Reiche–Kuhn sum rule is given by¹

$$\sum_{n'\ell'm'} f_{n'\ell'm',n\ell m} = Z, \quad (4.4)$$

where Z is the number of electrons in the atom and the sum implicitly includes the accessible continua. When applied to Na, this sum rule leads to $\sum f_{n'\ell'm',n\ell m} = 11$,

which tells us nothing about the oscillator strength of transitions of the outer electron alone. More useful sum rules, valid for one electron in a central potential, are¹

$$\sum_{n'} \bar{f}_{n'\ell-1,n\ell} = -\frac{1}{3} \frac{\ell(2\ell-1)}{2\ell+1} \quad (4.5)$$

and

$$\sum_n \bar{f}_{n'\ell+1,n\ell} = \frac{1}{3} \frac{(\ell+1)(2\ell+3)}{2\ell+1}, \quad (4.6)$$

from which it is apparent that

$$\sum_{n'\ell'} \bar{f}_{n'\ell',n\ell} = 1, \quad (4.7)$$

where $\ell' = \ell \pm 1$. Eqs. (4.5)–(4.7) apply to, for example, the transitions of the valence electron of Na, in spectral regions where the transitions of the Na^+ core are unimportant. Since the lowest lying excited states of Na^+ lie approximately 25 eV above the ground state of Na^+ , and 30 eV above the ground state of Na, Eqs. (4.5)–(4.7) can be used to describe the transitions of the valence electron of Na due to photons of energy less than 10 eV.

The attractions of the oscillator strengths are that they are dimensionless and they sum to 1. Thus the determination of one oscillator strength is often enough to give a good indication of the overall distribution of oscillator strengths. We shall see that the effect of nonzero quantum defects is to redistribute the alkali oscillator strengths relative to those of H. If we return to Eqs. (4.5) and (4.6) we can see that the strongest $\ell \rightarrow \ell-1$ transitions are to lower energies and that the strongest $\ell \rightarrow \ell+1$ transitions are to higher energies.

Finally, we introduce the Einstein A coefficient, $A_{n'\ell',n\ell}$, which defines the spontaneous decay rate of the $n\ell$ state to the lower lying $n'\ell'$ state. Explicitly,¹

$$A_{n'\ell',n\ell} = \frac{4e^2\omega_{n\ell,n'\ell'}^3}{3\hbar c^3} \frac{\ell_{\max}}{2\ell+1} |\langle n'\ell'|r|n\ell\rangle|^2. \quad (4.8)$$

In terms of the average oscillator strength we can write the A coefficient as

$$A_{n'\ell',n\ell} = \frac{-2e^2\omega_{n\ell,n'\ell'}^2}{\hbar c^3} \bar{f}_{n'\ell',n\ell}. \quad (4.9)$$

The minus sign comes from the fact that $\bar{f}_{n'\ell',n\ell} < 0$. The radiative lifetime, $\tau_{n\ell}$, of the $n\ell$ state is simply the inverse of the total radiative decay rate, which is obtained by summing $A_{n'\ell',n\ell}$ over all lower lying $n'\ell'$ states. Explicitly

$$\tau_{n\ell} = \left[\sum_{n'\ell'} A_{n'\ell',n\ell} \right]^{-1}. \quad (4.10)$$

From Eq. (4.8) we see that the A values contain a factor of ω^3 , which generally means that the transition with the highest frequency contributes the most to the radiative decay rate and therefore dominates the overall dependence on n .

For all Rydberg $n\ell$ states save the s states the highest frequency decay transition is to the lowest lying state of $\ell - 1$. For an s state it is the transition to the lowest p state. In either case, for a high $n\ell$ state, as $n \rightarrow \infty$ the frequency of the highest frequency transition approaches a constant. Thus, in the limit of high n , the A value depends only on the radial matrix element between the Rydberg state and the low lying state. Only the part of the Rydberg state wavefunction which spatially overlaps the wavefunction of the low lying state contributes to the matrix element in Eq. (4.8). As a result, the squared radial matrix element of Eq. (4.8) exhibits an n^{-3} scaling due to the normalization of the Rydberg state wavefunction. Correspondingly,

$$\tau_{n\ell} \propto n^3. \quad (4.11)$$

For $\ell \approx n$ states the same reasoning does not apply. For the $\ell = n - 1$ state the only allowed transition is to the $n' = n - 1$, $\ell' = n - 2$ state. In this case the frequency of the transition is not constant but is given by $1/n^3$. When cubed, this frequency contributes an n^{-9} scaling to the decay rate. Offsetting this scaling is the fact that the radial matrix element represents the size of the atom in the n and $n - 1$ states. Since $\langle n - 1 \ n - 2 | r | n \ n - 1 \rangle \sim n^2$ the lifetimes of the highest ℓ states scale as n^5 , i.e.

$$\tau_{n\ell=n-1} \propto n^5. \quad (4.12)$$

The average lifetime for a statistical mixture of ℓm states of the same n is given by^{1,2}

$$\bar{\tau}_n \propto n^{4.5}. \quad (4.13)$$

Finally, it is useful to extend the definition of oscillator strength above the ionization limit using

$$\frac{d\bar{f}_{\varepsilon'\ell',n\ell}}{dW} = \frac{2}{3} \omega_{\varepsilon'\ell',n\ell} \frac{\ell_{\max}}{2\ell+1} |\langle \varepsilon\ell' | r | n\ell \rangle|^2, \quad (4.14)$$

where the continuum wavefunction is normalized per unit energy.

The photoionization cross section of a statistical mixture of m levels of the $n\ell$ state to the $\varepsilon'\ell'$ continuum is then given by

$$\sigma = \frac{2\pi^2}{c} \cdot \frac{d\bar{f}_{\varepsilon'\ell',n\ell}}{cdW}. \quad (4.15)$$

It is interesting to examine the dependence on n and ℓ of the oscillator strength and lifetimes for the H and alkali atoms. First we show in Fig. 4.1 a plot of the oscillator strengths of the H $1s \rightarrow np$ transitions using the format of Fano and Cooper.³ Above the ionization limit $d\bar{f}/dW$ is plotted directly. Below the ionization limit the oscillator strength to the np level is plotted as a rectangular block of area $\bar{f}_{np,1s}$ from $W = -1/2(n - 1/2)^2$ to $W = -1/2(n + 1/2)^2$, i.e. it is approximately $1/n^3$ wide, $\bar{f}_{np,1s} n^3$ high, and centered at energy $W = -1/2n^2$. Thus the area of each block corresponds to the oscillator strength for the excitation of one np state. Plotting the oscillator strength in this fashion brings out clearly the fact that

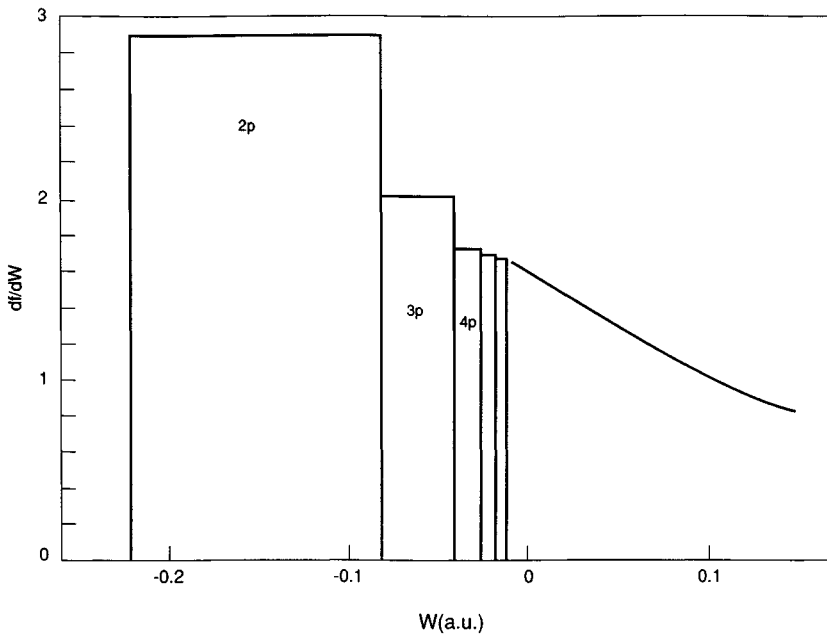


Fig. 4.1 Oscillator strength distribution from the H 1s state to the excited p states. Below the ionization limit the area of each block corresponds to the average oscillator strength from the 1s state to an np state.

oscillator strength per unit energy passes smoothly from the bound to the continuous region of the spectrum. From Fig. 4.1 it is also apparent that at the limit the oscillator strength is slowly varying, and thus $\bar{f}_{np,1s} \propto n^{-3}$ as expected from the normalization of the np wavefunctions near the origin. Fig. 4.1 and direct calculation show that a reasonable fraction, about one half, of the oscillator strength lies above the ionization limit.

In the excited states of H the lifetimes exhibit a predictable behavior. For states of a given n the p states decay most rapidly, followed by the d, f, g, s, h . . . states in that order.⁴ Putting aside the s states, the dominant decay mechanism for each $n\ell$ series is to the lowest $n'\ell - 1$ state. The frequency dependence of this transition upon ℓ leads to the observed ℓ ordering of the lifetimes. The ns states have long lifetimes due to the fact that the radial matrix elements to the lower lying p states are small. For the high ns states of H the first two lobes of the ns radial wavefunction overlap the 2p wavefunction, and their contributions cancel to some extent. In contrast, only one lobe, the innermost, of the nd radial wavefunction overlaps the 2p radial wavefunction. The nd wavefunction also has a larger amplitude than the ns wavefunction at the position of the 2p wavefunction because of the centrifugal potential, which ensures that the nd atom has less kinetic energy in radial motion. In general, matrix elements involving decreases in n accompanied by a decrease in ℓ by one are larger than matrix elements involving

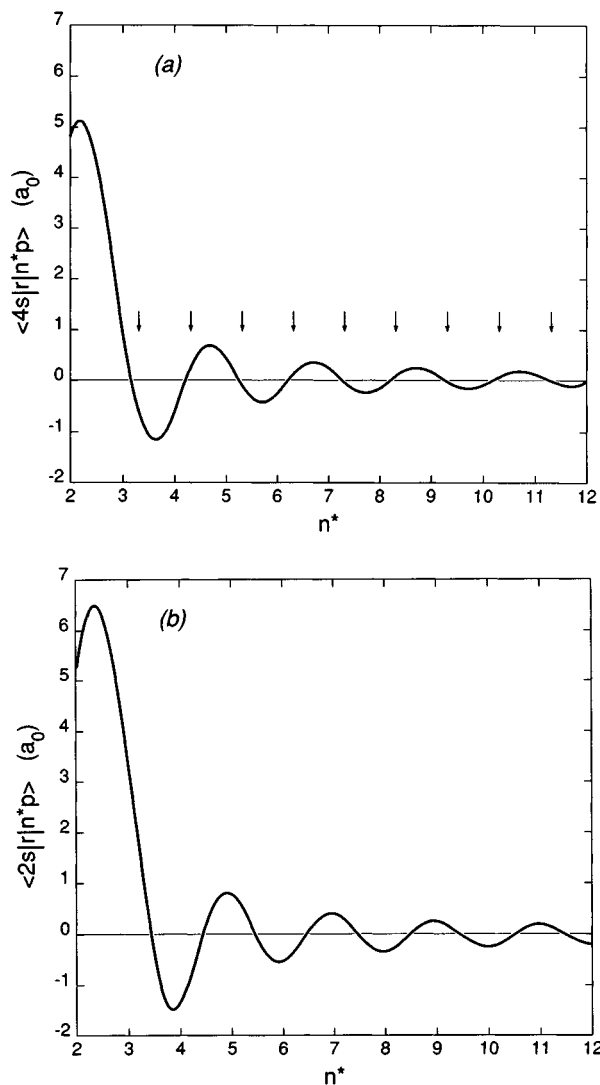


Fig. 4.2 (a) Radial matrix element of the K $4s \rightarrow n^*p$ transition with n^* a continuous variable. The quantum defect of the $4s$ state is 2.23. The quantum defects of the np states are 1.71, so the np states fall at the locations shown by the arrows near where the matrix element crosses zero. (b) H $2s \rightarrow n^*p$ radial matrix element as a function of n^* . Note that the maximum amplitudes of the matrix element occur at integer values of n^* .

decreases in n with an increase of ℓ by one. With the exception of the ns states, all states decay predominantly by $\Delta\ell = -1$ transitions.^{1,4}

Now let us consider alkali atoms. We begin by considering the K $4s \rightarrow np$ excitations analogous to the excitation from the ground state of H. In K the situation is in fact quite different, as shown by Fig. 4.2(a), a plot of the radial

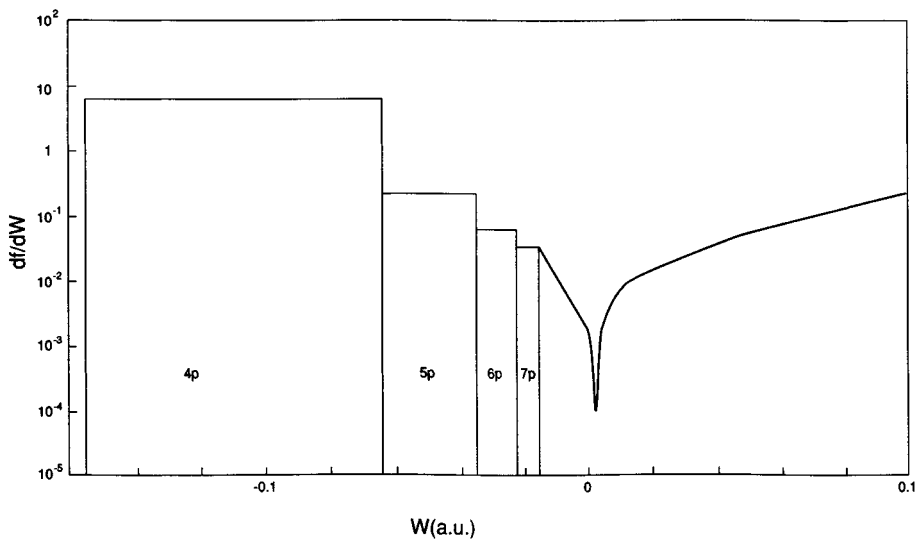


Fig. 4.3 Oscillator strength distribution from the K 4s state to the np states and ϵp continuum. Note that the vertical scale is logarithmic, not linear as in Fig. 4.1. The oscillator strengths to the high np states are orders of magnitude smaller than in H.

matrix element $\langle n^*p|r|4s \rangle$ for K with n^* a continuous variable. The K 4s state has a quantum defect of 2.23, and the high lying p states have quantum defects of 1.71, so that $n^* = 0.29(\text{mod } 1)$. In Fig. 4.2 we have shown the locations of the np states. It is quite apparent that the $4s \rightarrow 4p$ transition has a large matrix element, but the matrix elements to higher n states fall near the zero crossings in the matrix element and are therefore very small. The analogous plot of the matrix element between the H 2s and np states is shown in Fig. 4.2(b). The H $\langle 2p|r|2s \rangle$ matrix element is larger than the K $4s \rightarrow 4p$ matrix element due to the smaller binding energy of the H 2s state, and all the matrix elements between the 2s and the higher np states are further away from the zero crossings and hence larger than the K $\langle np|r|4s \rangle$ matrix elements.

If we convert the squares of the matrix elements shown in Fig. 4.2 to average oscillator strengths, by multiplying by $2\omega/3$, we find that in H the $2s \rightarrow 2p$ transition has vanishing oscillator strength due to the $2s \rightarrow 2p$ degeneracy. All the oscillator strength must come from the $n \geq 3$ $2s \rightarrow np$ and $2s \rightarrow \epsilon p$ transitions. On the other hand, in K the $4s \rightarrow 4p$ transition has a significant oscillator strength since the 4p state lies halfway between the 4s state and the ionization limit. In fact, it has an oscillator strength of very nearly 1, consuming nearly all of the available K $4s \rightarrow np$ oscillator strength. In Fig. 4.3 we show a plot of $d\bar{f}/dW$ for K. The values for the bound oscillator strengths and for the continuum more than 0.03 au above the limit are taken from Marr and Creek.^{5,6} The values of $d\bar{f}/dW$ near the minimum just above the limit are taken from the relative measurements of Sandner *et al.*⁷ normalized to the bound state oscillator strengths given by Marr

and Creek.⁵ In K the variation in $d\bar{f}/dW$ is so great that Fig. 4.3 is plotted on a logarithmic scale. As shown, $d\bar{f}/dW$ declines throughout the bound spectrum reaching a minimum just above the ionization limit. This minimum, usually termed the Cooper minimum, is found in the continua of all the alkalis save Li.

The origin of the minimum is most easily understood by considering a spinless K atom. In a spinless K atom the $4s$ – np radial matrix elements decrease with increasing energy, as implied by Fig. 4.3. The decrease with increasing energy of the np state occurs because the nodes of the bound np , or continuum ϵp , wavefunction become increasingly close together with increasing energy. Stated another way, the $4s$ – np or $4s$ – ϵp radial matrix element is an integral of the product of a fixed function, $r^3 R_{4s}(r)$, and a function the wavelength of which decreases with energy, $R_{np}(r)$ or $R_{\epsilon p}(r)$. In general the resulting integral oscillates about zero as a function of energy. In K the first change in sign is just above the ionization limit, leading to $d\bar{f}/dW = 0$ and a vanishing photoionization cross section. Having neglected spin we predict a zero cross section instead of the nonzero minimum of Fig. 4.3. The energy at which the minimum occurs is determined by the radial phases of the ground state and ϵp wavefunctions. In Na, K, Rb, and Cs these phases differ by $\sim \pi/2$, since the ground state and np quantum defects differ by $1/2$. For these atoms the Cooper minimum occurs in the continuum not far above the ionization limit. In Li the quantum defects of the ground state and the np states differ by $1/4$, and $d\bar{f}/dW$ has its minimum below the ionization limit.

As shown by Fig. 4.3, the cross section does not actually vanish, but only decreases to a nonzero minimum. As pointed out by Seaton,⁸ when the spin orbit interaction is taken into account there are two ϵp continuum waves, $\epsilon p_{1/2}$ and $\epsilon p_{3/2}$, which have slightly different radial phases corresponding to the slightly different quantum defects of the bound $np_{1/2}$ and $np_{3/2}$ series. As a result of having different phases the $\epsilon p_{1/2}$ and $\epsilon p_{3/2}$ radial matrix elements cross zero at slightly different energies. Each of these continuum channels has its own oscillator strength, and the total cross section is never zero. While the location of the minimum is a function of the difference in the quantum defects of the ground state and the excited np states, the minimum cross section and the width of the minimum are proportional to the np fine structure splitting, or the difference in the $np_{1/2}$ and $np_{3/2}$ quantum defects. Consequently the minimum cross section and the width of the Cooper minimum increase from Na to Cs.

Since the radial matrix elements from the ground state to the $np_{1/2}$ and $np_{3/2}$ states are not the same, due to the differing radial phases, the oscillator strengths and intensity ratios of the transitions to the $np_{1/2}$ and $np_{3/2}$ states cannot be in the 1:2 ratio predicted by the angular factors.⁸ The largest deviations occur in Cs, for the Cs $6s_{1/2}$ – $\epsilon p_{1/2}$ matrix element vanishes just above the ionization limit.⁹ As a result, just below the ionization limit large intensity ratios have been measured.^{10–13} For example, at $n=30$ Raimond *et al.* measured a ratio of 1:1170.¹³ The measurements agree with calculations of Norcross¹⁴ which predict that the

intensity ratio at $n=\infty$ should be roughly 1:12,000. While the Cooper minimum in K falls much closer to the ionization limit than it does in Cs, both the $4s\text{--}\epsilon p_{1/2}$ and $4s\text{--}\epsilon p_{3/2}$ matrix elements cross zero at nearly the same energy, and in the bound states the maximum $np_{1/2}\text{:}np_{3/2}$ intensity ratio is only 1:4.¹⁵

Radiative lifetimes

Both the absorption cross sections and the lifetimes of low ℓ states of alkali atoms are different from the hydrogenic values. The most glaring examples occur the alkali np states. In H the np states have very short radiative lifetimes, but in an alkali atom they have very long lifetimes, due to their small oscillator strengths to the ground state. For example, as shown in Figs. 4.1 and 4.3, the oscillator strengths from the ground states of H and K to the high, $n \approx 10$, np states differ by a factor of 100. This factor, when multiplied by the ratio of the squares of the frequencies, 16, leads to K $np \rightarrow 4s$ Einstein A coefficients roughly 1000 times smaller than their H $np \rightarrow 1s$ counterparts. The K np lifetimes are not a thousand times longer than the H np lifetimes because decay to states other than the ground state plays a significant role in the total decay rate.

The radiative lifetimes of many excited states of Li,¹⁶ Na,^{17–21} K,²² Rb,^{23–25} and Cs,^{26–28} have been measured using a variety of techniques, the most common being time resolved laser induced fluorescence, which is typically carried out using a cell, as shown in Fig. 4.4. In all cases, the observed lifetimes are in reasonable agreement with values calculated in the coulomb approximation, corrected for the decrease due to black body radiation.²⁹ In the following chapter we show that if the 0 K lifetime of a state is τ that its lifetime at a finite temperature T is given by^{24,30}

$$\frac{1}{\tau^T} = \frac{1}{\tau} + \frac{1}{\tau^{bb}}. \quad (4.16)$$

where $1/\tau^{bb}$ is the black body radiation induced decay rate.

Systematic lifetime measurements have been made for Rb^{23–25} and Na.^{17–21} In both cases the lifetimes of the s, p, d, and f states have been measured, and we discuss the Na lifetimes as a representative example. The lifetimes of Na $n < 15$ s, p, d, and f states have been measured using the time and wavelength resolved laser induced fluorescence arrangement shown in Fig. 4.4. Na atoms at a pressure of 10^{-6} Torr in a pyrex cell are excited from ground state to the 3p state and then to the ns or nd states using two dye laser pulses of 5 ns duration. The ns and nd lifetimes are easily measured by observing the time resolved $ns \rightarrow 3p$ or $nd \rightarrow 3p$ fluorescence with a photomultiplier^{17,18}. These fluorescent decays have large branching ratios and are easily detected in spite of the fact that they are at the

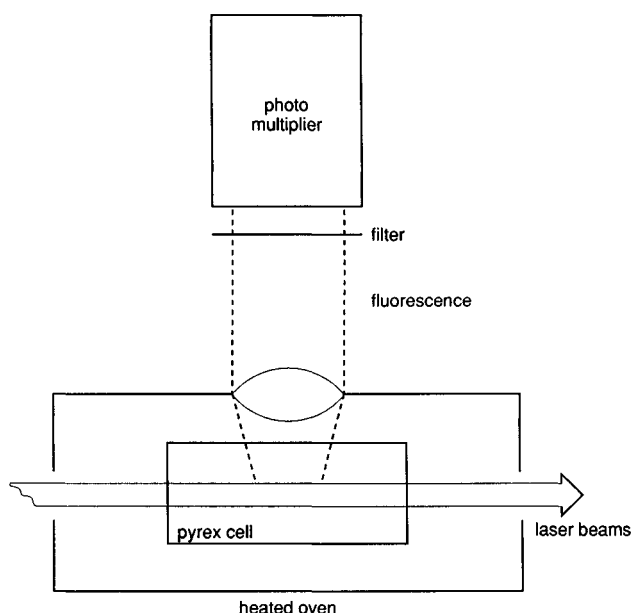


Fig. 4.4 Schematic diagram of a typical arrangement for laser induced fluorescence measurements of lifetimes. A pulsed laser beam (or beams) passes through a heated glass cell containing alkali vapor and the time and wavelength resolved fluorescence is detected at a right angle.

same wavelength as the second exciting laser. For rapidly decaying low lying states the fluorescence is typically stronger than the scattered laser light, and for slowly decaying high lying states the phototube can be gated on after the laser pulse. The longer lived Na np lifetimes can be observed by exciting the nd state which quickly decays, leaving an observable fraction, $\sim 10\%$, of the initial nd population in the much longer lived np state.¹⁹ The time resolved $4p \rightarrow 3s$ fluorescence can be directly detected, but for higher np states the $np \rightarrow 3s$ fluorescence has too short a wavelength to pass through a glass cell. Instead, the time resolved $ns \rightarrow 3p$ fluorescence, the second step of the $np \rightarrow ns \rightarrow 3p$ cascade is observed. The long time behavior of the observed signal reflects the lifetime of the np state, which is much longer than the lifetime of either the ns or nd states. Finally, the lifetimes of several Na nf states have been measured using a resonant microwave field to equalize the populations in the nd and nf levels.²⁰ Observing the time resolved fluorescence from either the nf or nd state gives a decay rate indistinguishable from the nd state decay rate, indicating that the nf lifetime is very nearly the same as the nd lifetime. Had the nf lifetimes been much longer than the nd lifetimes, this method would have been relatively insensitive to the nf lifetimes.

The fluorescence methods described above work well for states of principal quantum numbers less than 15. For higher principal quantum numbers they do not work so well for three reasons. First, the amplitude of the fluorescence signal

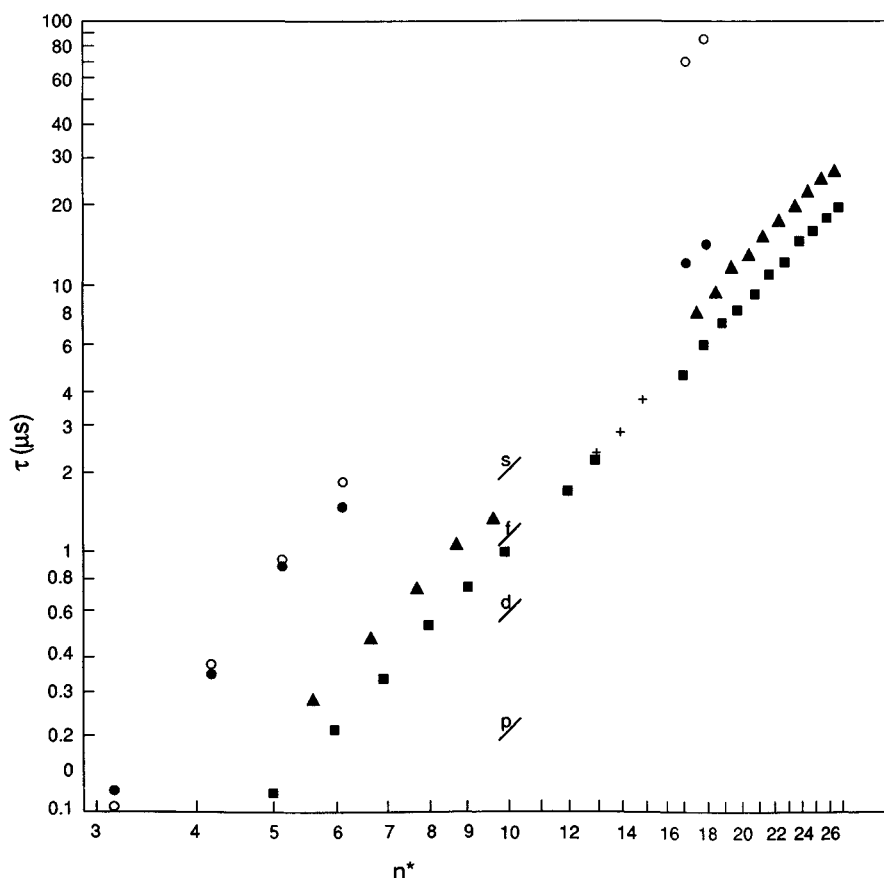


Fig. 4.5 Na radiative lifetimes vs n^* . Experimental values for ns ($\blacktriangle$), np ($\circ$), nd ($\blacksquare$) and nf ($+$) states are shown. The lifetimes states below $n = 15$ have been measured by fluorescence techniques, at temperatures of approximately 400 K. The lifetimes of $n > 15$ states have been measured by field ionization, the ns and nd states at 30 K and the np states at 300 K. The theoretical 0 K np lifetimes are also shown ($\circ$). They are far above the measured values at high n due to black body radiation. Finally, the hydrogenic lifetimes are shown by the line segments ($-$) (from refs. 4, 18–22, 30).

decreases as n^{-6} , second, the atoms pass out of the field of view due to their longer lifetimes, and, third, black body radiation can appreciably alter the lifetimes.⁸ The time delayed field ionization method employed by Spencer *et al.*²¹ to measure the Na ns and nd lifetimes circumvents all of these problems. They passed a well collimated Na atomic beam through an interaction region cooled to 30 K. The atoms were excited by two pulsed dye laser beams collinear with the atomic beam and tuned to the $3s \rightarrow 3p$ and $3p \rightarrow ns$ or $3p \rightarrow nd$ transitions. At a variable time after the laser excitation a field ionization pulse was applied to atoms in the interaction region and the resulting ions detected. The atoms which passed out of the interaction region between the laser pulse and the field ionization pulse were

Table 4.1. *Lifetime parameters for alkali atoms.*

		s	p	d	f
Li ^a	τ_0 (ns)	1.39	5.69	0.59	0.96
	α	2.80	2.78	2.92	3.06
Na ^b	τ_0 (ns)	1.38	8.35	0.96	1.13
	α	3.00	3.11	2.99	2.96
K ^b	τ_0 (ns)	1.32	6.78	5.94	0.83
	α	3.00	2.78	2.82	2.95
Rb ^b	τ_0 (ns)	1.43	2.76	2.09	0.76
	α	2.94	3.02	2.85	2.95
Cs ^b	τ_0 (ns)	1.43	4.42	0.96	0.69
	α	2.96	2.94	2.93	2.94

^a(computed from values in ref. 29)^b(from ref. 31)

replaced by atoms which had been excited further upstream by the laser. Using this approach they were able to make measurements at delay times of up to 25 μ s, corresponding to about 2 cm of travel of the atoms. It is useful to note that if the interaction region is cooled the only criterion for the field ionization pulse is that it be strong enough to ionize the Rydberg state in question. If significant black body redistribution occurs, as it does at 300 K, the field ionization pulse must be chosen with more care, as described in Chapter 5.

In Fig. 4.5 we show measured values of Na *ns*, *np*, *nd*, and *nf* lifetimes vs n^{*3} on logarithmic scales. The scales differ by a factor of 3, so an n^{*3} dependence appears as a 45° line. As shown by Fig. 4.5 the n^{*3} dependence suggested by Eq. (4.11) is obtained. The values for $n \leq 15$ have all been obtained using fluorescence detection from atoms in a cell at about 425 K. For all but the *np* states the black body radiation is a negligible effect. For the *np* states we also show the calculated 0 K values of Theodosiou, and for the 7*p* state, the measured value is about 30% below the calculated 0 K value and consistent with the calculated 425 K value of Theodosiou.²⁹ For $n > 15$ all values have been obtained using field ionization. The *ns* and *nd* values are those obtained by Spencer *et al.*²¹ at 30 K and are clearly in reasonably good agreement with an extrapolation of the values obtained at lower n using fluorescence detection. For the 18*p* and 19*p* lifetimes measured at 300 K, however, black body radiation reduces the lifetimes far below an n^{*3} extrapolation of the lifetimes of the lower *np* states.³⁰

We also show in Fig. 4.5 short 45° lines at $n = 10$ corresponding to the lifetimes of the hydrogenic *ns*, *np*, *nd*, and *nf* series. The Na *np* and *nd* lifetimes are respectively factors of 50 and 2 longer than their hydrogenic counterparts, while the Na *ns* lifetimes are about 30% shorter than the H *ns* lifetimes. The Na and H *nf* lifetimes are the same, which is not surprising as the Na *nf* states and the Na *nd* states to which they predominantly decay all have quantum defects of nearly zero.

The lifetimes of alkali Rydberg states can be calculated accurately, as shown by Theodosiou,²⁹ who has made extensive comparisons between observed lifetimes and calculated values. From this work it is also apparent that the black body radiation correction of Eq. (4.16) is important for high lying states. In light of these two points, it seems that the most compact and consistent way of presenting the alkali lifetimes is by a fit of the calculated 0 K values to the form³¹

$$\tau = \tau_0(n^*)^\alpha \quad . \quad (4.17)$$

The values of τ_0 and α are presented in Table 4.1. As shown by Table 4.1, the values of α are all near 3, as expected from Eq. (4.11). The biggest discrepancies occur in the Li and K np series which have their Cooper minima nearest the ionization limit. It is also apparent that the np lifetimes are the longest, as expected from Figs. 4.1 and 4.2. In using the values of Table 4.1 it is worth remembering that for high n states the black body decay rate is ℓ independent, so the lifetimes of the longest lived states are the most affected by black body radiation.³²

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